

The midline of a triangle and of a trapezium

Two theorems with one idea: join the midpoints and you get a segment parallel to the base, of exactly the average length.

LESSON 15

1 × 40 MIN

GEOMETRY 8, ТЕМЫ 10–11

STAGE 9 · 13.2

LESSON CLOCK

4'

13'

8'

13'

2'

Warm-up	4 min
Explanation	13 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State and prove the midline theorem for a triangle.
- State and use the midline theorem for a trapezium.
- See the triangle case as the trapezium case with $b = 0$.
- Use midlines to solve length problems and to prove other results.

TEXTBOOK

National	Темы 10–11, pp. 26–28
Cambridge	Learner’s Book pp. 281–284
Workbook	Workbook 13.2

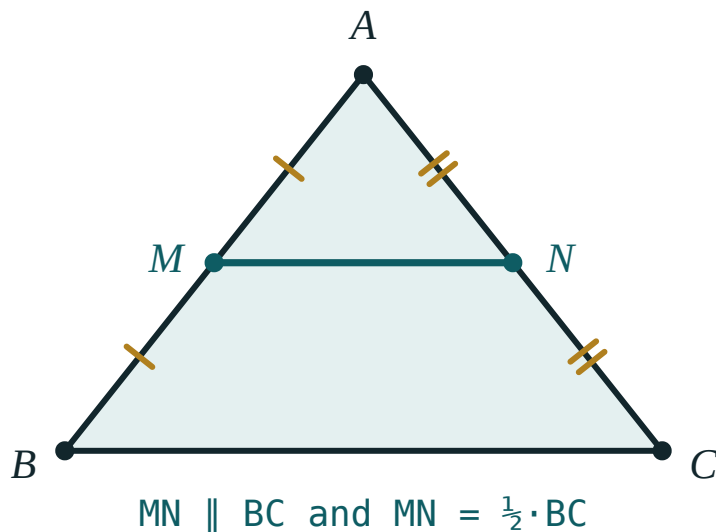
01

Explanation

The midline of a triangle

DEFINITION

The **midline** of a triangle is the segment joining the midpoints of two of its sides.



M and N are the midpoints of AB and AC. The midline MN is parallel to BC and exactly half its length.

THEOREM

The midline of a triangle is parallel to the third side and equal to half of it:

$$MN \parallel BC \text{ and } MN = \frac{1}{2} BC$$

Proof. Extend MN beyond N to a point P with $NP = MN$. Then $\triangle ANM \cong \triangle CNP$ by SAS ($AN = NC$, $MN = NP$, vertically opposite angles at N). Hence $CP = AM = MB$ and $\angle CPN = \angle AMN$, which are alternate angles — so $CP \parallel MB$. Now $MBCP$ has one pair of sides equal and parallel, so by Test 1 it is a parallelogram. Therefore $MP \parallel BC$ and $MP = BC$; and since $MN = \frac{1}{2} MP$, the theorem follows. ■

CONSEQUENCE

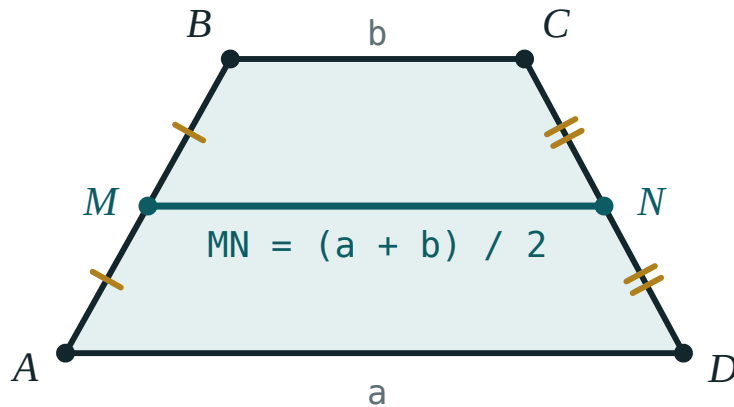
The three midlines of a triangle cut it into four congruent triangles, each similar to the original with half the side lengths and a quarter of the area. The middle triangle has perimeter equal to half the original.

The midline of a trapezium

THEOREM

The midline of a trapezium — the segment joining the midpoints of the two legs — is parallel to the bases and equal to their half-sum:

$$MN \parallel AD \parallel BC \quad \text{and} \quad MN = \frac{a + b}{2}$$



The midline of a trapezium is the average of the two bases.

Proof. Draw the diagonal BD and let it meet MN at K . In $\triangle ABD$, MK joins the midpoints of AB and BD — a midline, so $MK = \frac{1}{2}AD = \frac{a}{2}$. In $\triangle BCD$, KN is a midline, so $KN = \frac{1}{2}BC = \frac{b}{2}$. Adding gives $MN = \frac{a + b}{2}$. ■

ONE THEOREM, NOT TWO

Let the shorter base shrink to a point, $b = 0$. The trapezium becomes a triangle and the formula becomes $MN = \frac{a}{2}$ — the triangle midline theorem. The triangle case is the trapezium case with one base of length zero.

Using midlines

Two habits that solve most midline problems:

- When you see **two midpoints**, join them — you have created a midline.
- When you see **one midpoint** and want a second, draw a parallel to a side; the midpoint corollary from the last lesson supplies it.

Area is also useful: the area of a trapezium is $S = MN \cdot h$ — the midline multiplied by the height.

02 Worked examples

EXAMPLE 1 In $\triangle ABC$, $BC = 18\text{ cm}$. M and N are the midpoints of AB and AC . Find MN .

① MN joins two midpoints — it is a midline.

② $MN = \frac{1}{2} BC$

Midline theorem.

③ $MN = 9\text{ cm}$

ANSWER

$MN = 9\text{ cm}$

EXAMPLE 2 A trapezium has bases 8 cm and 14 cm and height 5 cm . Find its midline and area.

① $MN = \frac{8 + 14}{2} = 11\text{ cm}$

Half-sum of the bases.

② $S = MN \cdot h = 11 \cdot 5$

Area of a trapezium.

③ $S = 55\text{ cm}^2$

ANSWER

$MN = 11\text{ cm}, S = 55\text{ cm}^2$

EXAMPLE**3**

The midline of a trapezium is 12 cm and one base is 3 cm shorter than the other. Find both bases.

① $a + b = 2 \cdot 12 = 24$

From the midline formula.

② $a = b + 3$

The given relation.

③ $b + 3 + b = 24 \implies 2b = 21$

④ $b = 10.5, a = 13.5$

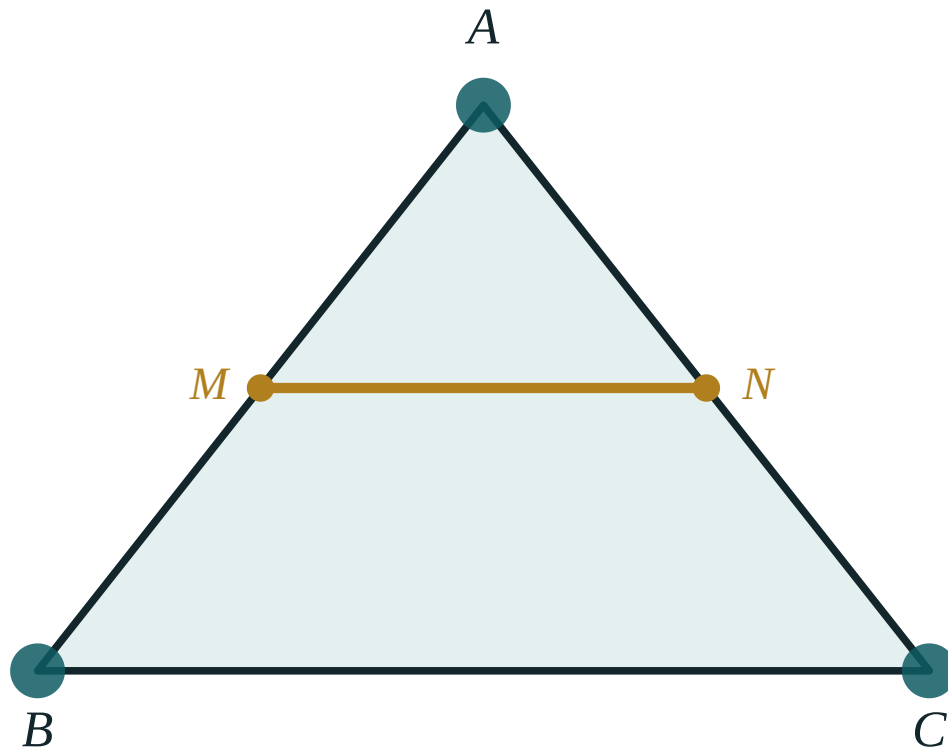
ANSWER

10.5 cm and 13.5 cm

03 Interactive model

Drag a vertex to a wild position — the ratio $BC : MN$ stays at 2.00 throughout.

The midline of a triangle



BC 260

MN 130

BC / MN 2

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Midline of a triangle	Uchburchakning o'rta chizig'i	Средняя линия треугольника
Midline of a trapezium	Trapetsiyaning o'rta chizig'i	Средняя линия трапеции
Midpoint	O'rta nuqta	Середина
Half-sum	Yarim yig'indi	Полусумма
Corollary	Natija	Следствие
Congruent triangles	Teng uchburchaklar	Равные треугольники
Parallel	Parallel	Параллельный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The midline of a triangle with base 20 cm is:

40 cm

10 cm

20 cm

5 cm

The midline of a trapezium with bases 7 and 13 is:

20

10

6

91

The three midlines of a triangle cut it into:

2 triangles

3 triangles

4 congruent triangles

6 triangles

A trapezium has midline 9 and height 6. Its area is:

54

27

15

108

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. In $\triangle ABC$, $BC = 18$ cm. Find the midline MN parallel to BC .

ANSWER 9 cm

2. A midline of a triangle is 7 cm. Find the side it is parallel to.

ANSWER 14 cm

3. A trapezium has bases 6 cm and 10 cm. Find the midline.

ANSWER 8 cm

4. A trapezium has bases 5 cm and 15 cm. Find the midline.

ANSWER 10 cm

5. A trapezium has midline 9 cm and height 6 cm. Find the area.

ANSWER 54 cm²

6. A triangle has sides 10, 12, 14. Find the perimeter of the triangle formed by its midlines.

ANSWER 18

7. The midline of a trapezium is 11 cm and one base is 8 cm. Find the other.

ANSWER 14 cm

Medium · the standard question

7 PROBLEMS

1. A trapezium has bases 8 cm and 14 cm and height 5 cm. Find the midline and area.

ANSWER 11 cm, 55 cm²

2. The midline of a trapezium is 12 cm and one base is 3 cm shorter than the other. Find both.

ANSWER 10.5 cm and 13.5 cm

3. In $\triangle ABC$ the midlines have lengths 4, 5, 6. Find the sides of $\triangle ABC$.

ANSWER 8, 10, 12

4. A trapezium has area 60 cm² and height 5 cm. Find its midline.

ANSWER 12 cm

5. The midline of a trapezium is 10 cm and the bases are in the ratio 2 : 3. Find them.

ANSWER 8 cm and 12 cm

6. In $\triangle ABC$, M and N are the midpoints of AB and AC, and MN = 6.5 cm. Find BC.

ANSWER 13 cm

7. A trapezium has bases 9 and 21 and legs 10 and 10. Find the midline and the height.

ANSWER Midline 15; $\frac{21 - 9}{2} = 6$, so $h = 8$

Hard · stretch and reason

7 PROBLEMS

1. *Prove the midline theorem for a triangle.*

ANSWER Extend MN to P with $NP = MN$; $\triangle ANM \cong \triangle CNP$ by SAS makes $MBCP$ a parallelogram, so $MP = BC$ and MN is half of it.

2. *Prove that the midline of a trapezium equals the half-sum of the bases.*

ANSWER The diagonal BD splits MN into two triangle midlines of lengths $\frac{a}{2}$ and $\frac{b}{2}$.

3. *The midlines of a triangle cut it into four triangles. Compare their areas with the original.*

ANSWER Each is $\frac{1}{4}$ of the original area, and all four are congruent.

4. *The midline of a trapezium is 14 cm and the bases are in the ratio 3 : 4. Find both bases.*

ANSWER $a + b = 28$ in the ratio 3 : 4: 12 cm and 16 cm.

5. *P, Q, R, S are the midpoints of the sides of any quadrilateral. Prove $PQRS$ is a parallelogram.*

ANSWER PQ and SR are both midlines parallel to the diagonal AC and equal to half of it.

6. *The midline of a triangle divides it into a triangle and a trapezium. Compare their areas.*

ANSWER The small triangle is $\frac{1}{4}$ of the whole, so the trapezium is $\frac{3}{4}$ — a ratio of 1 : 3.

7. *A trapezium has midline 8 and the two bases differ by 4. Find the area if the height is 7.*

ANSWER Bases 6 and 10; $S = 8 \cdot 7 = 56$.

07 Homework

Homework — 5 problems

Geometry 8, Темы 10–11, pp. 26–28.

1. In $\triangle ABC$, $BC = 22$ cm. Find the midline parallel to BC .
2. A trapezium has bases 11 cm and 17 cm. Find the midline, then the area if the height is 6 cm.
3. The midline of a trapezium is 15 cm and one base is 4 cm longer than the other. Find both.

4. *The midlines of a triangle are 6, 7 and 8. Find the perimeter of the triangle.*
5. *Prove that the midline of a triangle is parallel to the third side.*